PROBLEM SOLVING

BACK

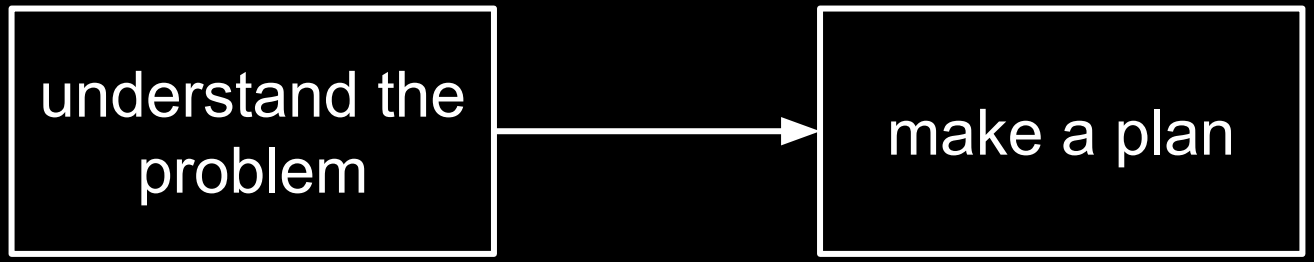
Pólya's approach to problem-solving

image source: http://doi.org/10.3932/ethz-a-000099441

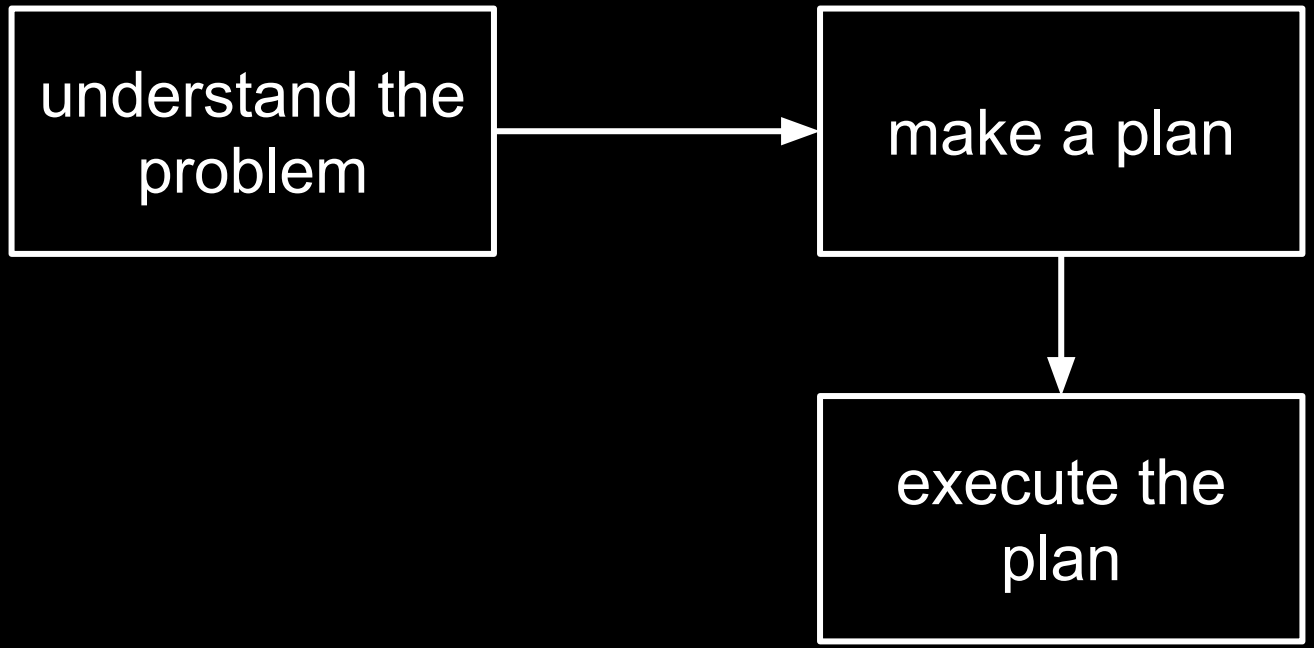
Pólya's approach to problem-solving

understand the problem

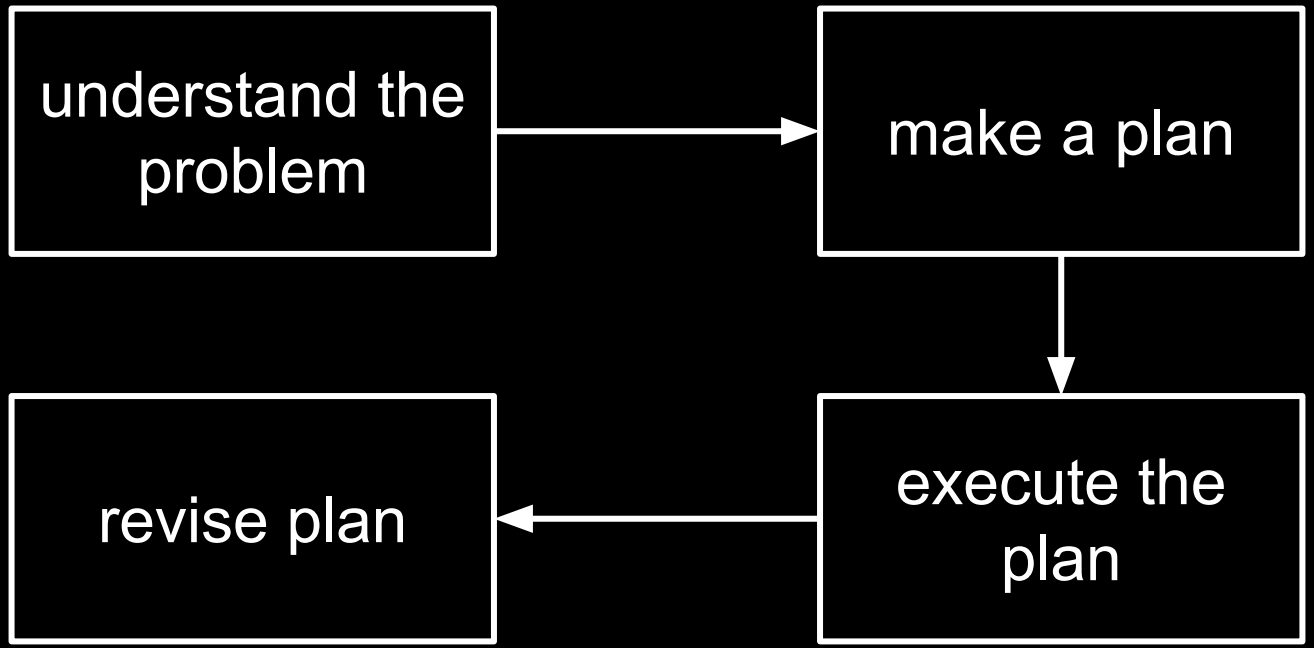
Pólya's approach to problem-solving



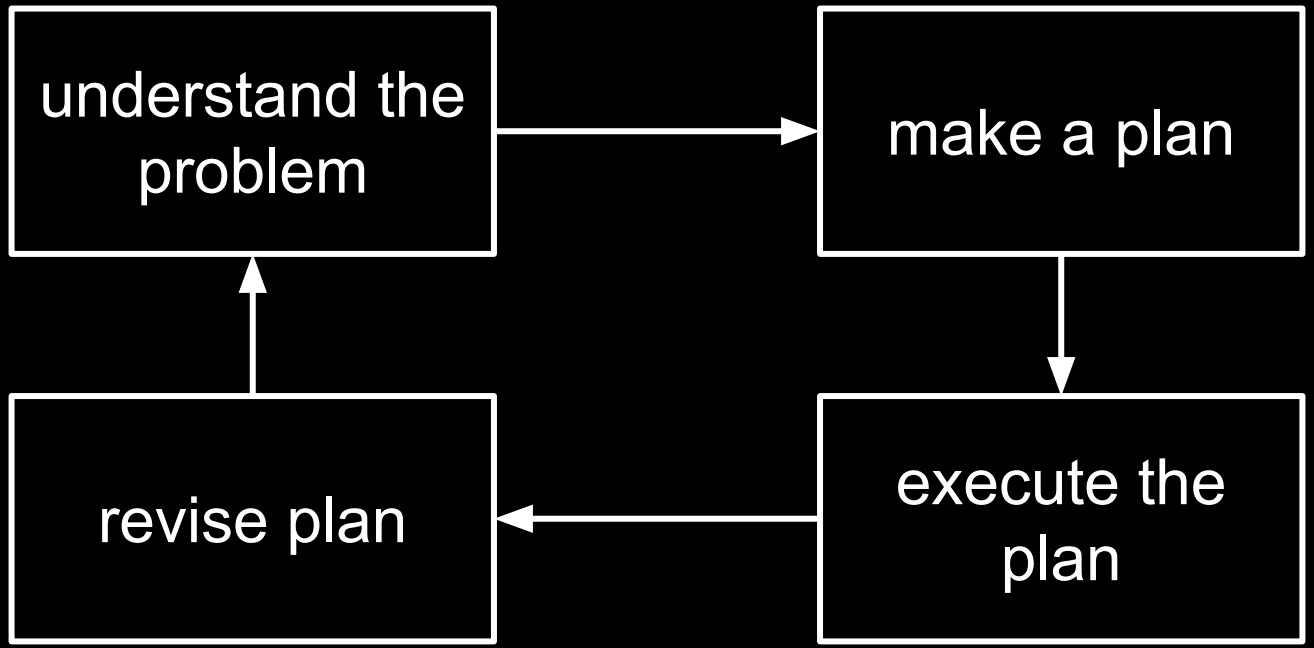
Pólya's approach to problem-solving



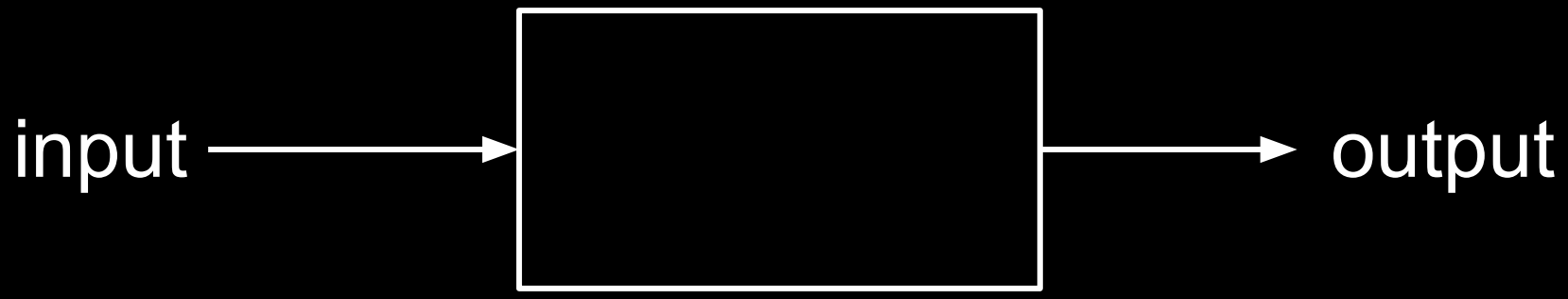
Pólya's approach to problem-solving



Pólya's approach to problem-solving

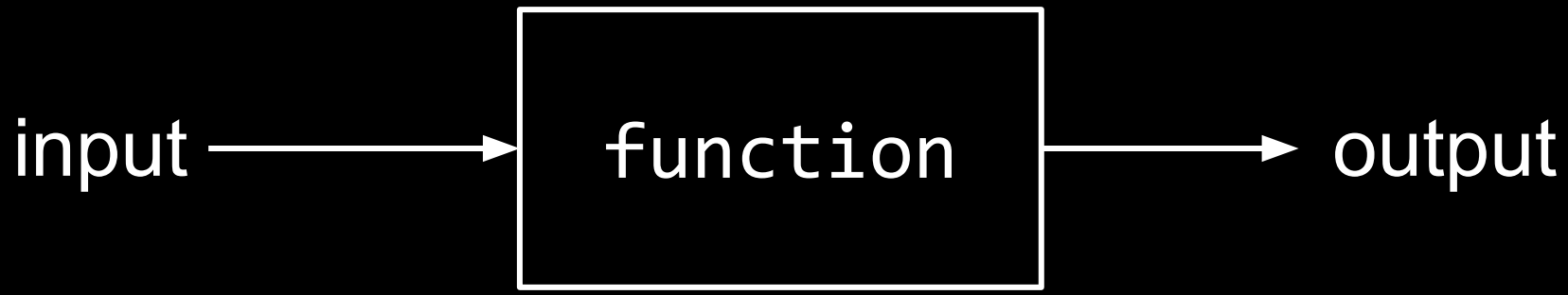


a model to represent problems

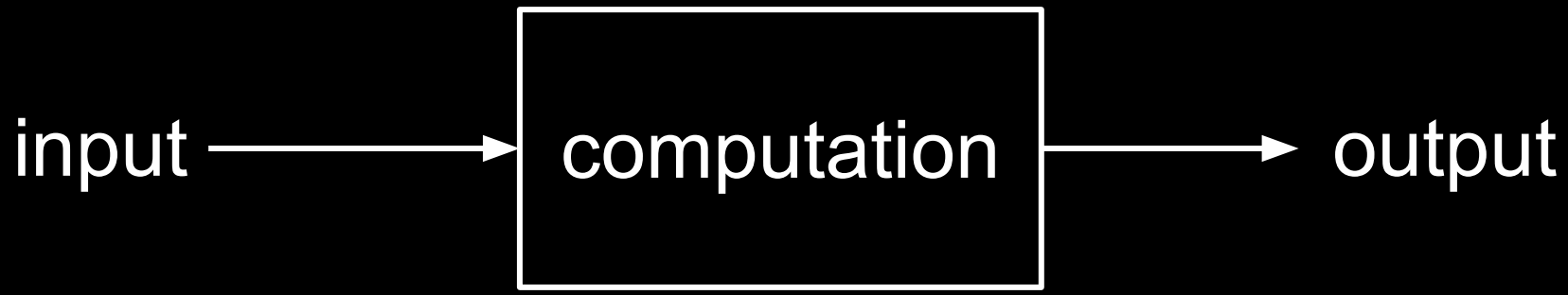


a model to represent problems

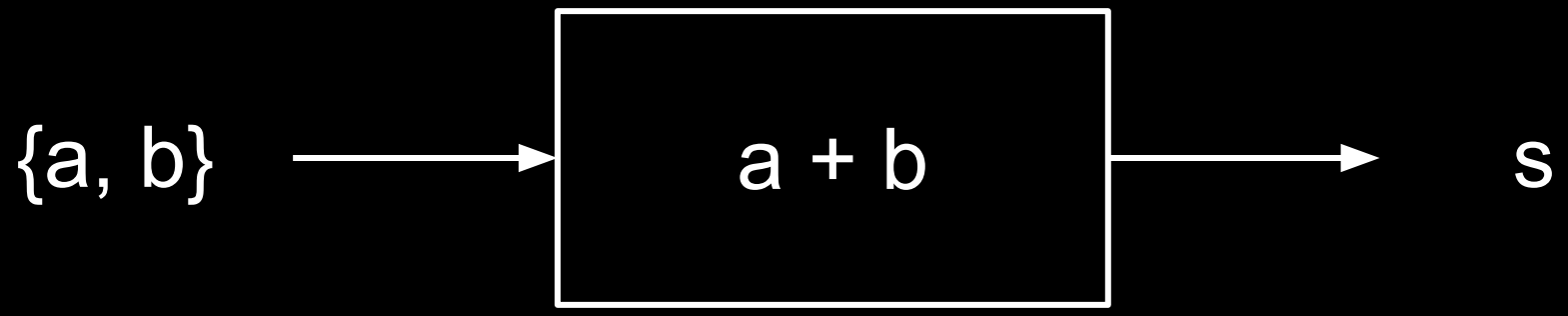
a model to represent problems

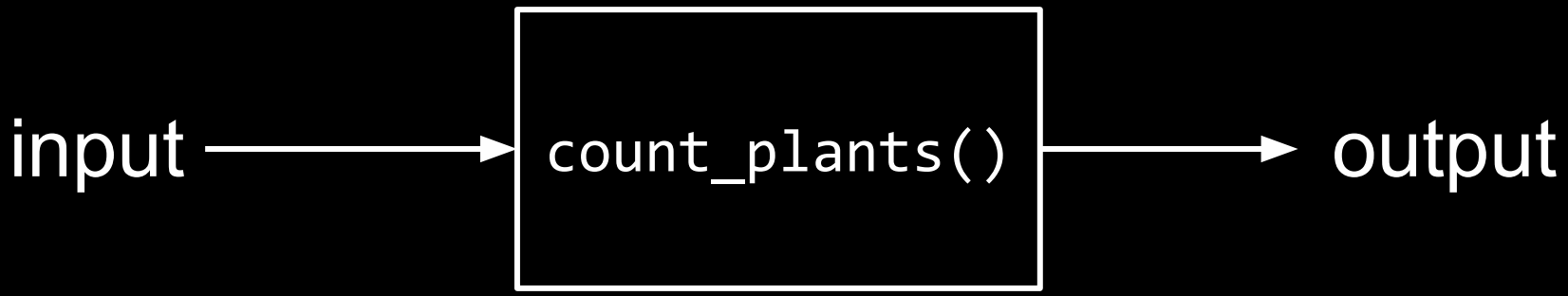


a model to represent problems



the rainbow experiment as an input - processing - output - problem


{a, b}
a + b
s

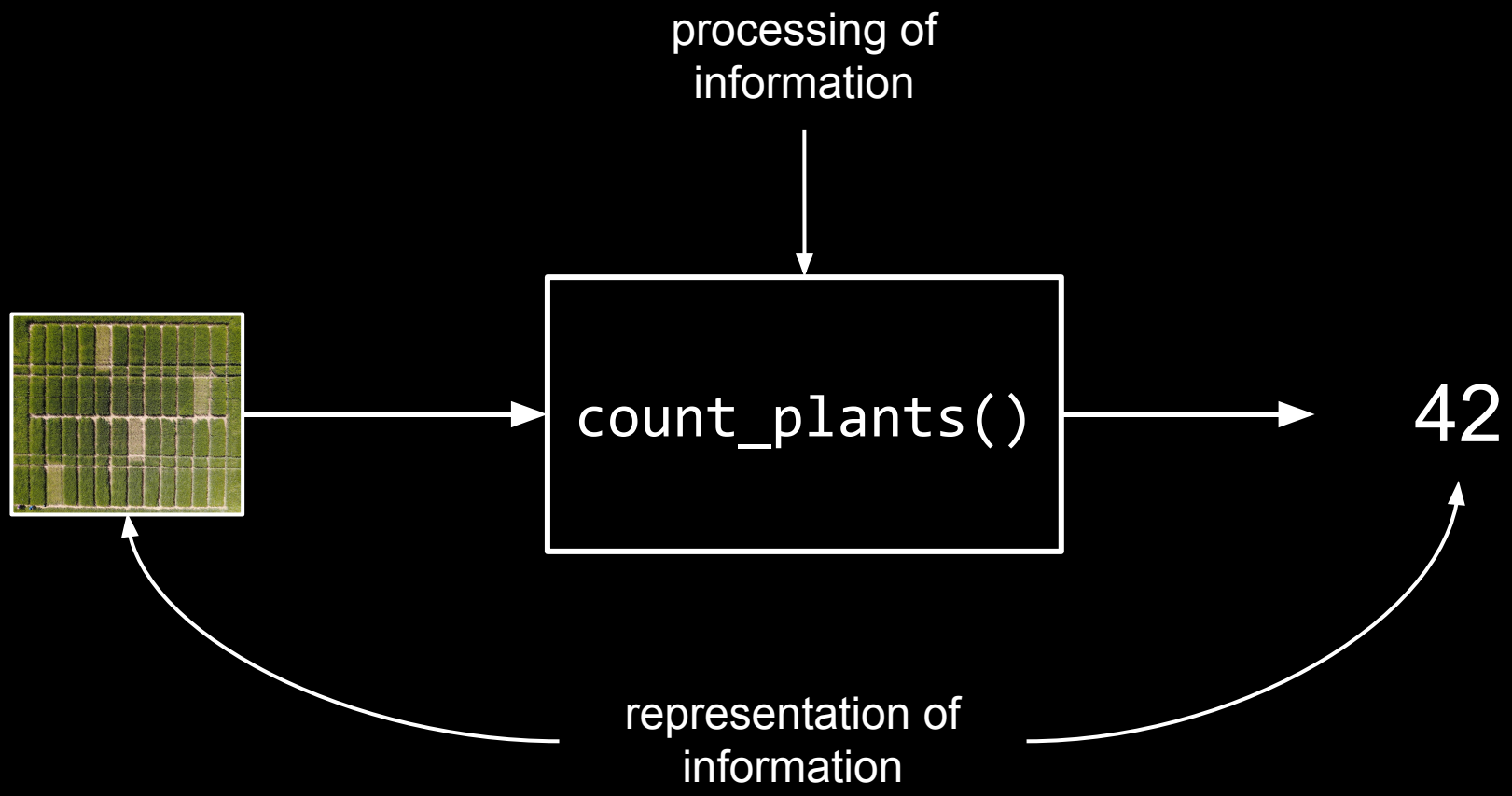

input
count_plants()
output

count_plants()

output

```
count_plants()
```

42


processing of information
count_plants()
42
representation of information

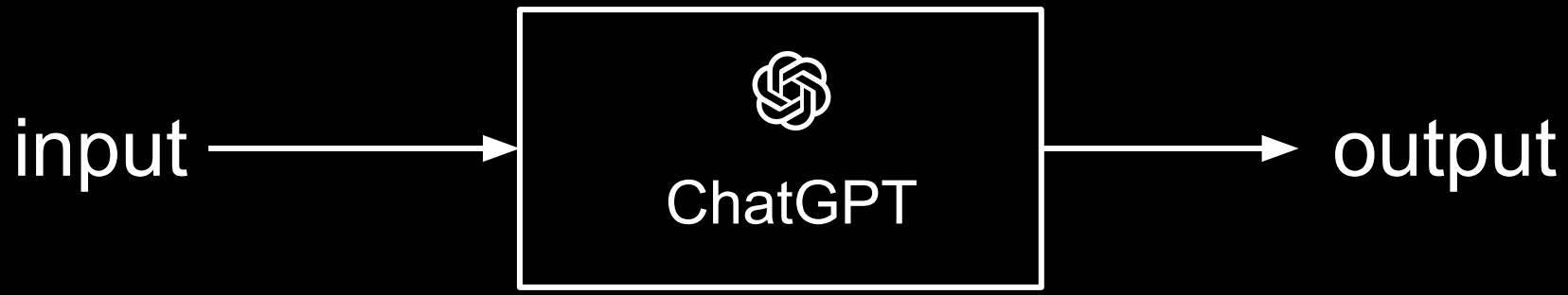
next_move()

E2 → E4

1: 0R	9: 0P	57: 1R
2: 0N	10: 0P	58: 1N
3: 0B	11: 0P	59: 1B
4: 0K	12: 0P	60: 1K
5: 0Q	13: 0P	61: 1Q
6: 0B	14: 0P	62: 1B
7: 0N	15: 0P	63: 1N
8: 0R	16: 0P	64: 1R

…

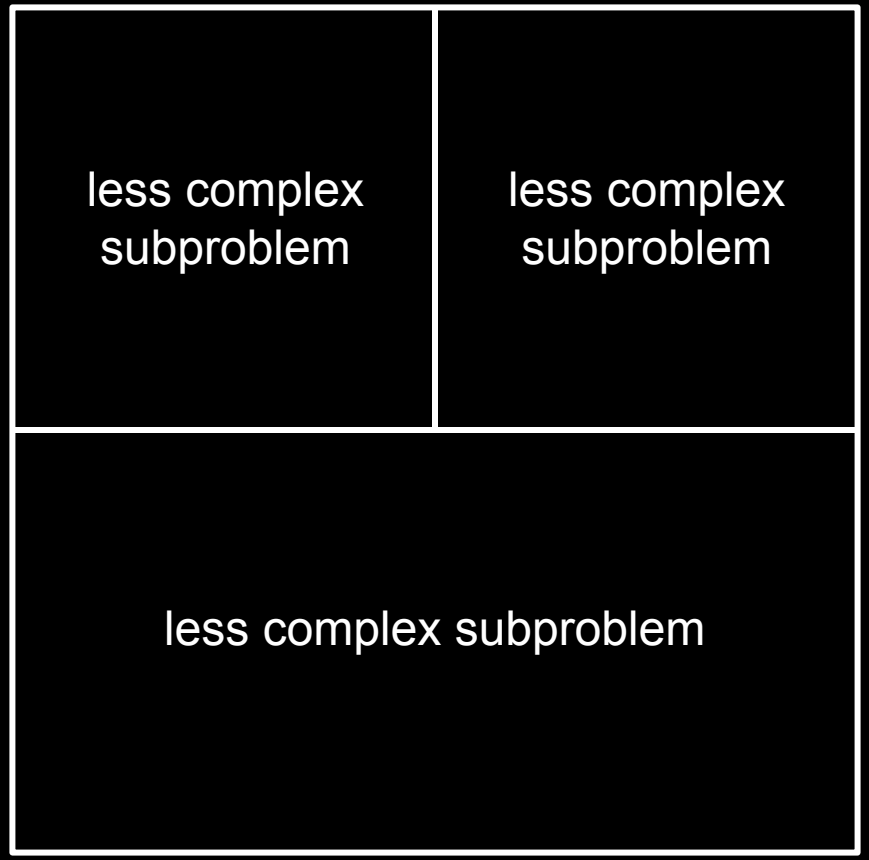
representation of information

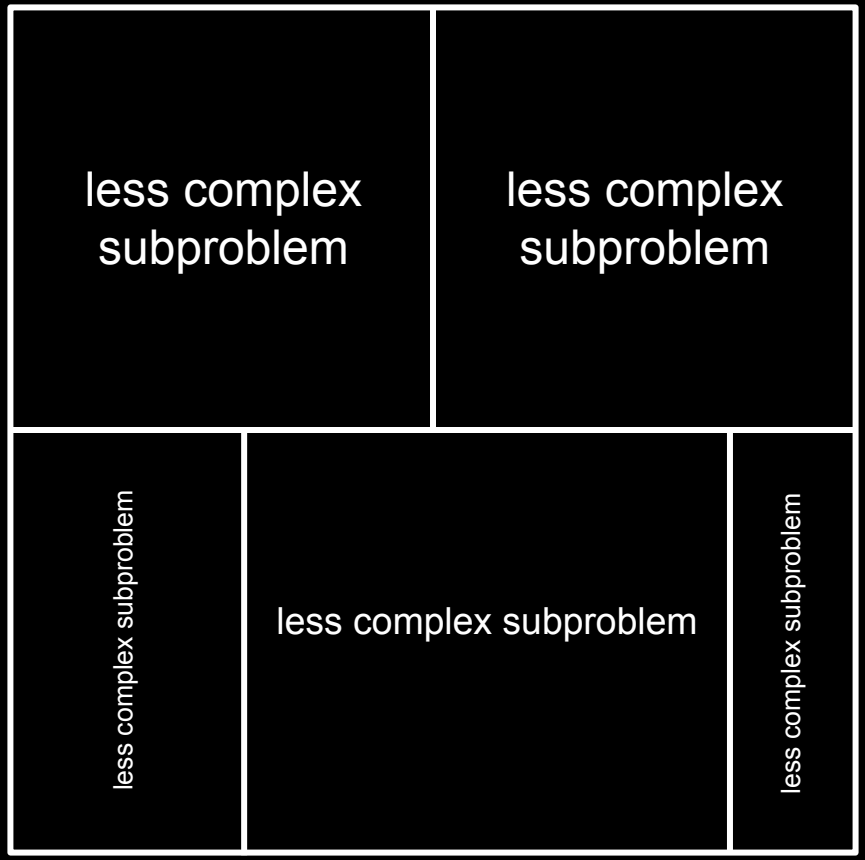

input
ChatGPT
output

problem solving strategies

problem decomposition

large and complex problem


less complex subproblem
less complex subproblem
less complex subproblem


less complex subproblem
less complex subproblem
less complex subproblem
less complex subproblem
less complex subproblem

divide and conquer

large and complex problem of type A

smaller problem of type A	smaller problem of type A
smaller problem of type A	smaller problem of type A

even smaller problem of type A	even smaller problem of type A
even smaller problem of type A	even smaller problem of type A
even smaller problem of type A	even smaller problem of type A
even smaller problem of type A	even smaller problem of type A

sorted list +
element
search()
yes / no

is 67 a prime number?

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

linear search

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

linear search

~~2~~, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

linear search

~~2~~, ~~3~~, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

linear search

~~2~~, ~~3~~, ~~5~~, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

linear search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~, ~~43~~, ~~47~~, ~~53~~, ~~59~~, ~~61~~, 67, 71, 73, 79, 83, 89, 97

⬆

19 steps… can’t we do better?

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
~~43~~, ~~47~~, ~~53~~, ~~59~~, ~~61~~, 67, 71, 73, 79, 83, 89, 97

⬆

large and complex problem

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

large and complex problem

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

smaller problem

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41

smaller problem

43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

binary search

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

binary search

67 != 41

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ~~41~~,
43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

binary search

67 > 41

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ~~41~~,
43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

binary search

67 > 41

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

67 != 71

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
43, 47, 53, 59, 61, 67, ~~71~~, 73, 79, 83, 89, 97

67 != 71

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
43, 47, 53, 59, 61, 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 < 71

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
43, 47, 53, ~~59~~, 61, 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 != 59

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
~~43~~, ~~47~~, ~~53~~, ~~59~~, 61, 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 > 59

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
~~43~~, ~~47~~, ~~53~~, ~~59~~, 61, 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 = 67

binary search

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
~~43~~, ~~47~~, ~~53~~, ~~59~~, ~~61,~~ 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 = 67

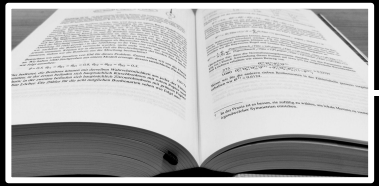
3 splits → much better

~~2~~, ~~3~~, ~~5~~, ~~7~~, ~~11~~, ~~13~~, ~~17~~, ~~19~~, ~~23~~, ~~29~~, ~~31~~, ~~37~~, ~~41~~,
~~43~~, ~~47~~, ~~53~~, ~~59~~, ~~61~~, 67, ~~71~~, ~~73~~, ~~79~~, ~~83~~, ~~89~~, ~~97~~

67 = 67

🤔

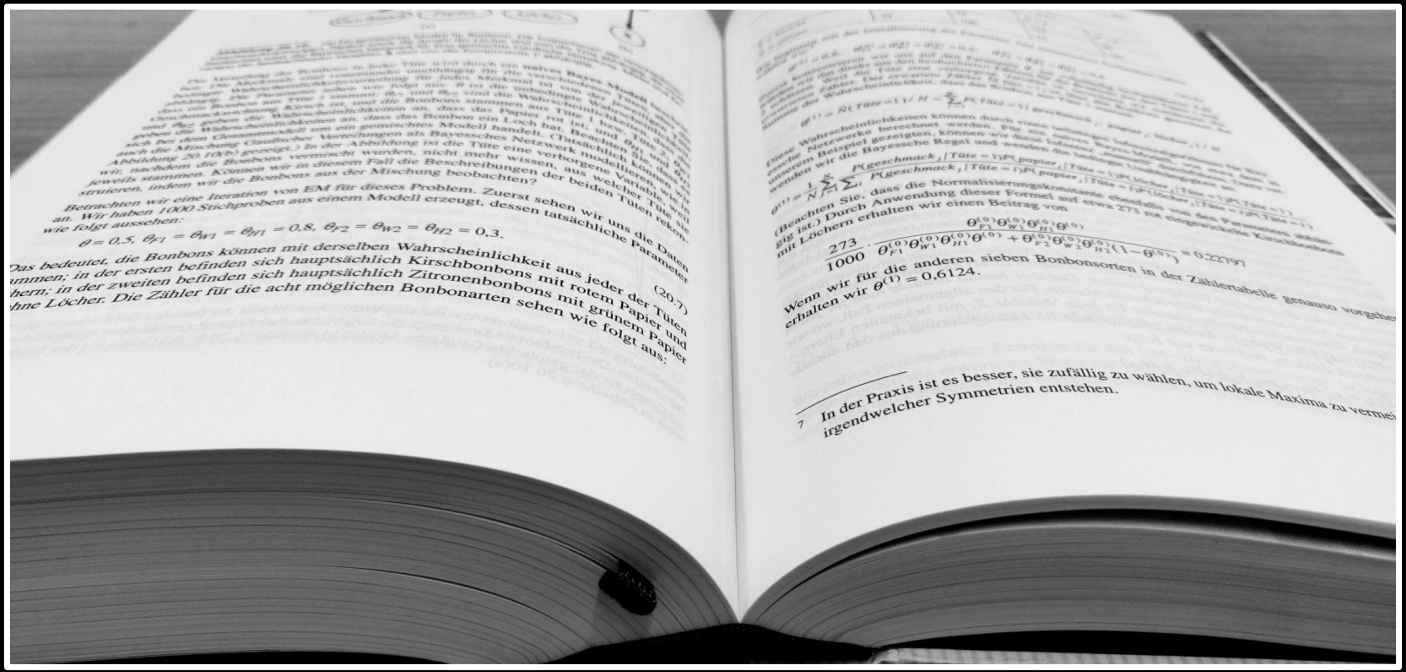
how efficient are linear and binary search in general?



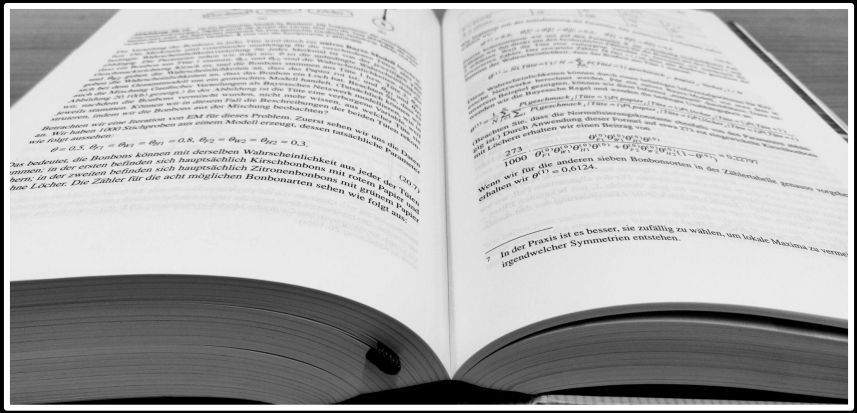
```
count_words()
```

word count

how many words are in the book?



strategies, anyone?



Beobachten wir eine Iteration von EM für dieses Problem. Zuerst sehen wir uns die Tabelle an. Wir haben 1000 Stichproben aus einem Modell erzeugt, dessen tatsächliche Parameter wie folgt aussehen:

$$\theta = 0.5,\ \theta_{F1} = \theta_{W1} = \theta_{H1} = 0.8,\ \theta_{F2} = \theta_{W2} = \theta_{H2} = 0.3. \tag{20.7}$$

Das bedeutet, die Bonbons können mit derselben Wahrscheinlichkeit aus jeder der Tüten kommen; in der ersten befinden sich hauptsächlich Kirschbonbons mit rotem Fett, Wrappern und Herzen; in der zweiten befinden sich hauptsächlich Zitronenbonbons mit grünem Fett, Wrappern und runden Löchern. Die Zähler für die acht möglichen Bonbonarten sehen wie folgt aus:

$$\hat{\theta}^{(1)} = \hat{\theta}(\mathbf{z}^{(1)}|\mathbf{x}) = 0.5 \sum_{j=1}^{N} P(F|x_j, z_j^{(1)}) \cdot \ldots$$

Unter Wahrscheinlichkeiten können durch die Anwendung der Formel für die bedingte Wahrscheinlichkeit berechnet werden, z. B. $P(F|x_j, z_j^{(1)}) = \ldots$. Wenn wir die Ergebnisse zusammenfassen, können wir sehen, dass die Bayesische Regel und unsere Formel für die Berechnung der Parameter θ identisch sind:

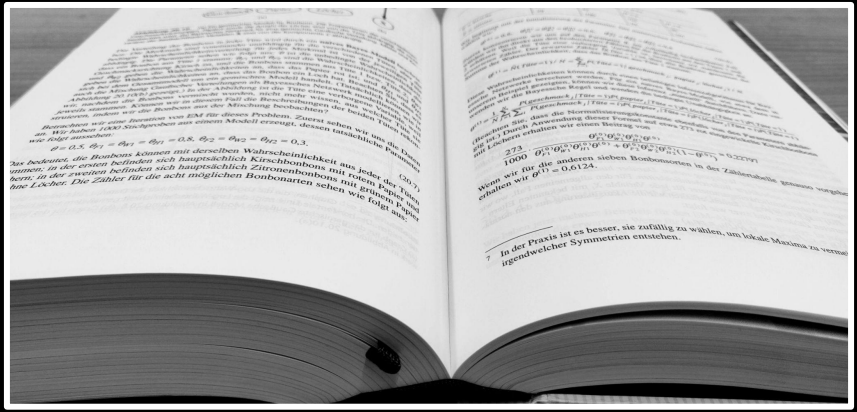
$$\theta^{(1)} = \frac{1}{N}\sum_{j=1}^{N} P(\text{Tüte} = 1|F, W, H) = \frac{1}{N}\sum_{j=1}^{N} P(\text{Tüte} = 1|x_j) = \ldots$$

Im nächsten Schritt sehen wir, dass die Normalisierungskonstante $\sum_{j=1}^{N} P(\text{Tüte} = 1|x_j)$ auf 273 fällt. Durch Anwendung dieser Formel mit einer 273 erhalten wir einen Beitrag von

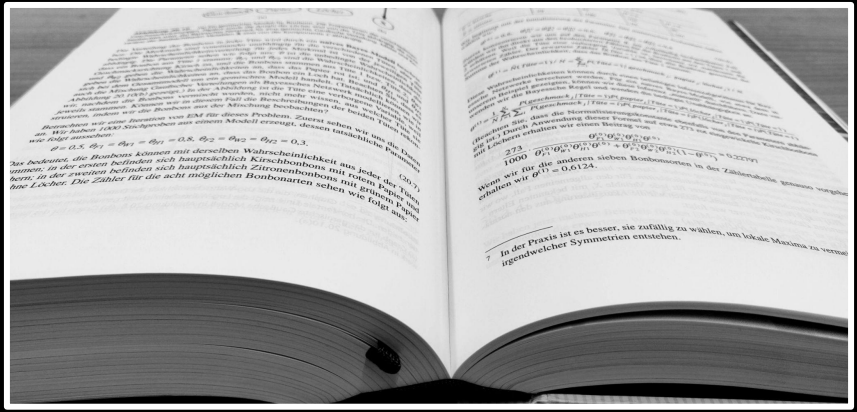
$$\frac{273}{1000}\ \theta_F^{x_F}\theta_W^{x_W}\theta_H^{x_H}\theta^{(1)} + \theta_F^{x_F}\theta_W^{x_W}\theta_H^{x_H}(1-\theta^{(1)}) = 0.2279$$

auf die Zähler für die anderen sieben Bonbonarten in der Zählertabelle gemäß vorgehen. Wenn wir für $\theta^{(1)} = 0{,}6124$ erhalten wir

7 In der Praxis ist es besser, sie zufällig zu wählen, um lokale Maxima zu vermeiden, ohne irgendwelcher Symmetrien entstehen.



page 1



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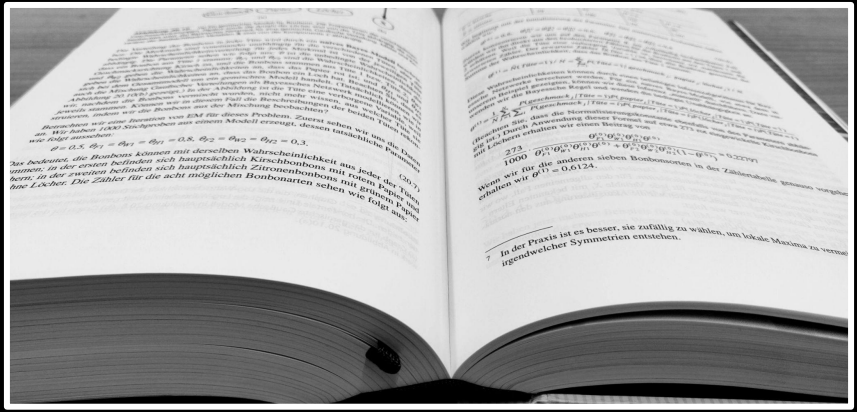
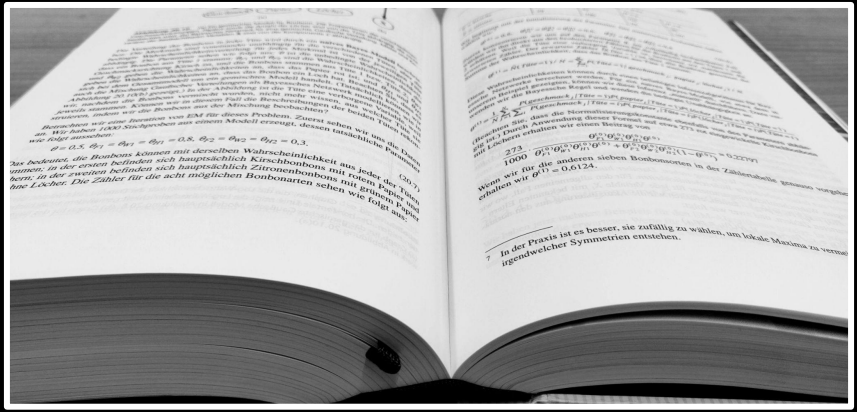
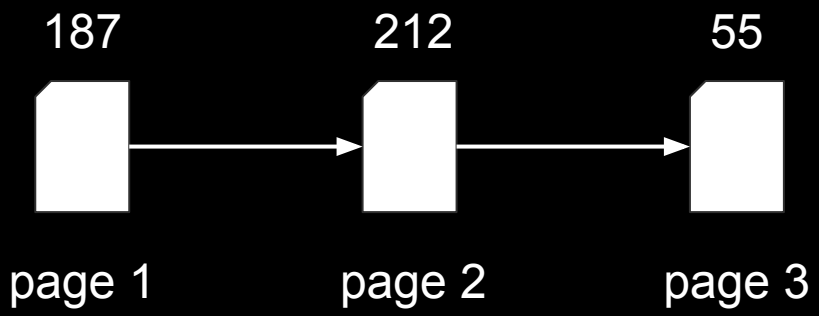
page 1



187 212

page 1 page 2

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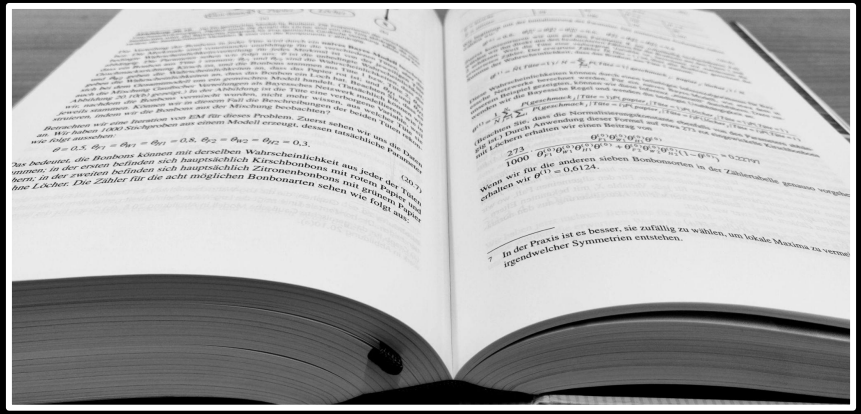
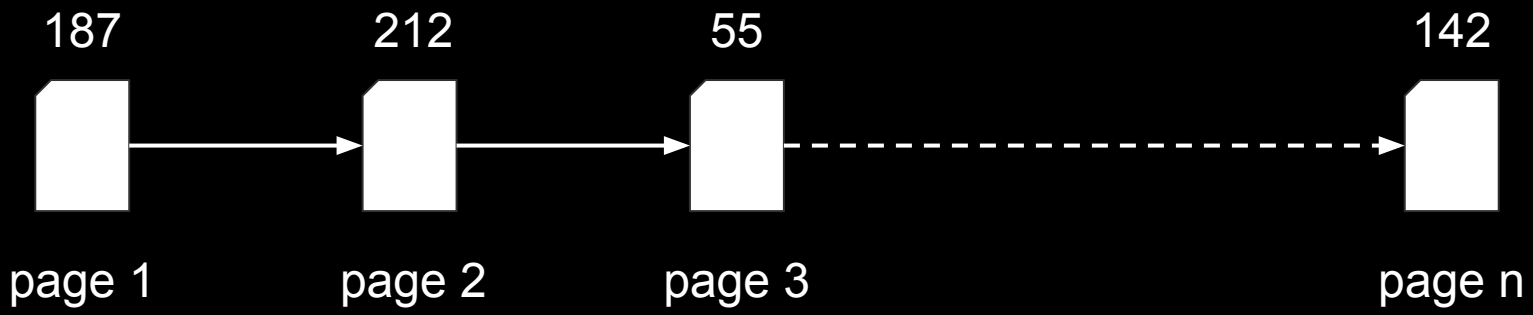
212
55
page 1
page 2
page 3

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page 1
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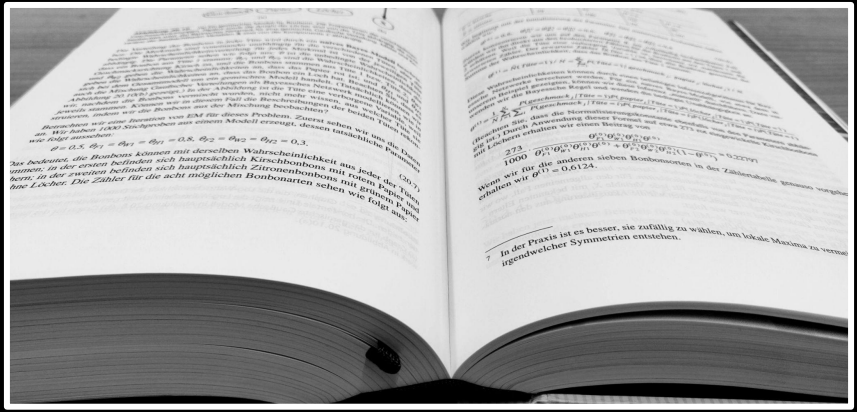


n = 1327 pages

Ø 2:23 minutes per page

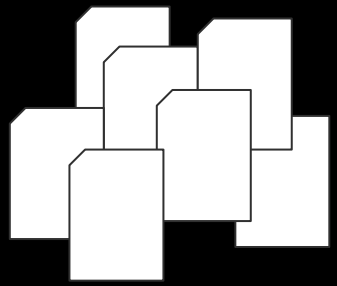
~ 52.34 hours

divide and conquer

+

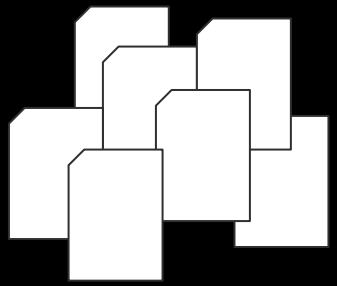
?

pages 1 - 700



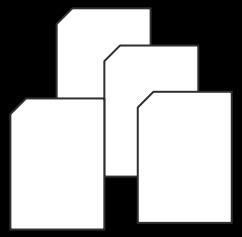
student 1

pages 701 - 1327



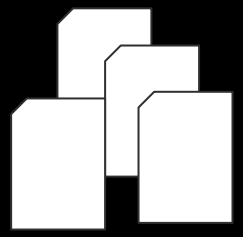
student 2

pages 1 - 350



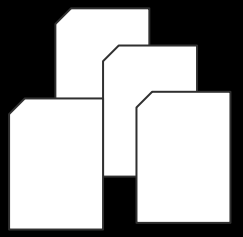
student 1

pages 351 - 700



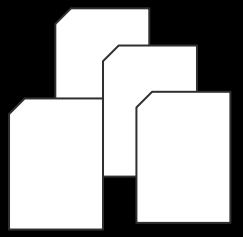
student 2

pages 701 - 1050



student 3

pages 1051- 1327

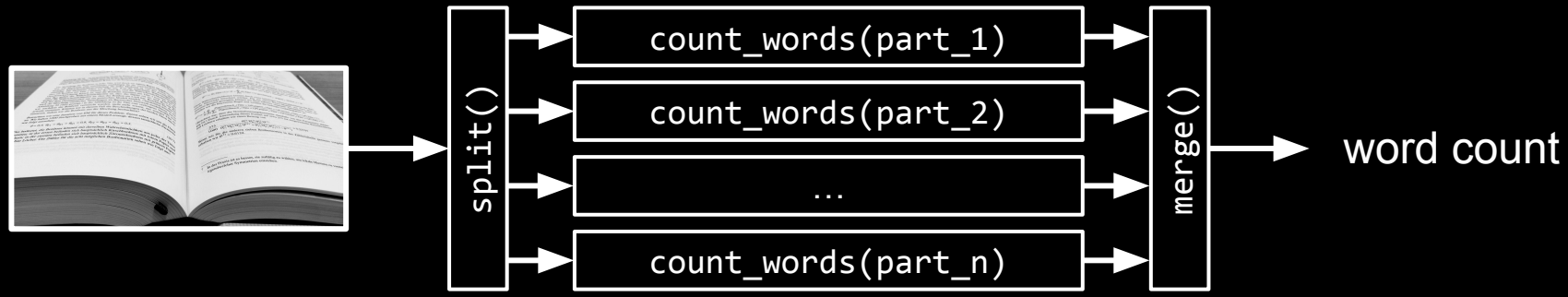


student 4

divide and conquer

+

distribution and parallelization


split()
count_words(part_1)
count_words(part_2)
...
count_words(part_n)
merge()
word count